

The lowest bracket sets the cap: an analytical fixed point for top wealth under a progressive wealth tax, with calibration to UK Sunday Times Rich List data

[Author]

(supervision and acknowledgements TBD)

Draft of June 1, 2026 for JEIC

Abstract

We present an agent-based model of an economy with seven occupationally heterogeneous agent classes and use it to derive an analytical characterisation of how the rates and thresholds of a progressive wealth tax interact with the capital-return rate r to determine top wealth in equilibrium. For a single rentier with wealth above a threshold θ , the fixed-point of the wealth dynamics under capital return r and marginal wealth-tax rate τ is $w^* = \tau\theta/(\tau - r)$, defined for $\tau > r$ and diverging for $\tau \leq r$. We validate this prediction against simulation and show that the empirical top wealth tracks the analytical fixed point quantitatively for $\tau > r$ at the lowest active threshold, with deviations attributable to secondary sinks (deposit pool, subsistence, inheritance tax) acting as an effective additional tax.

We then use the model to address the practical question of how much the conclusions of a wealth-tax design analysis depend on the assumed value of r . A partial calibration to the UK Sunday Times Rich List 2024 Pareto slope of -0.706 returns a best fit at $r = 0.06$, rent share 0.50, inheritance tax 0.40; the slope is reachable but the configuration produces approximately 26% population survival, well below UK reality. We document the Pareto frontier between matching the empirical tail and matching empirical mortality.

The principal substantive result is that the policy conclusions of the wealth-tax-design analysis depend critically on which r is adopted. At default $r = 0.04$ the Zucman flat 2% and Zucman progressive 2/4/8% schedules are not separable at $N = 10,000$, because no agent reaches the upper-tier thresholds. At calibrated $r = 0.06$ they separate cleanly: Zucman progressive produces zero billionaires while Zucman flat leaves 46. The upper tiers of a progressive schedule are mechanically idle at one parameter setting and load-bearing at the other. We report both, side by side, and discuss which conclusion should be considered the headline as a function of the assumed capital-return rate.

We engage Spangler and Sarkar [2022] as the closest published neighbour with a clean-toggle reproduction that holds bandwidth and class-bias intact and swaps only whether transmitted skill values are parent-derived or class-typical: parent-derived transmission raises Gini by 0.010 in our setting, a partial reproduction of their direction with a much smaller magnitude. We engage Benhabib et al. [2011] as the analytical neighbour for the fixed-point derivation.

A further result, which we believe is the contribution most likely to be policy-relevant, is that the wealth tax and an absolute cap on inheritance are *complementary instruments serving different goals*. The wealth tax raises substantial state revenue (Zucman 2% above £10M produces approximately £56B per year at simulated UK scale). The cap raises far less revenue but redistributes ownership directly: the wealth share held by the bottom 50% of the population rises from 0.0% under status quo to 19.2% under a £5M cap, and to 25.3% under cap-plus-tax. The two instruments compete for the upper-end tax base but otherwise stack cleanly. The choice between cap-alone, tax-alone, and the combination is a choice between three different policy objectives, not between three policies aimed at the same one. Cultural-mechanism results are reported separately in a companion submission targeting Artificial Life; the regime-classification and Sankey results are reported in a companion

submission targeting JASSS; the policy-discourse implications of the cap result are reported in a companion submission targeting Fiscal Studies.

1 Introduction

The wealth-tax debate has been refreshed by the 2024 Zucman [2024] G20 proposal of a 2% minimum effective tax on the assets of ultra-high-net-worth individuals. Within the existing literature the question of which structural feature of a wealth tax does the work (the threshold, the rate, the progressivity, the spending package, the inheritance regime) is partially answered by analytical optimal-tax theory [Piketty and Saez, 2013, Benhabib et al., 2011] and by kinetic-tax ABMs [Bertotti and Modanese, 2018, Cardoso et al., 2025]. This paper contributes to that conversation by deriving a simple analytical fixed-point characterisation of the top wealth under a flat marginal wealth tax, validating it against an agent-based simulation, and showing that the policy conclusions of the simulation depend sensitively on the assumed value of the capital-return rate r .

We use a stylised stock-flow ABM with seven occupationally typed agents (workers, makers, employers, lenders, rentiers, speculators, and a single state). The model is documented in detail in our JASSS companion submission and in the project repository; we recap the essentials in Section 3. The contribution this paper foregrounds is the wealth-tax-design analysis, the analytical fixed point, and the calibration sensitivity.

We organise the paper around four results.

First, the analytical fixed-point characterisation (Section 4). For a rentier with wealth above a threshold θ , accruing capital return r and paying a marginal wealth tax τ , the dynamics have a finite fixed point $w^* = \tau\theta/(\tau - r)$ when $\tau > r$ and diverge when $\tau \leq r$. This is a first-order rigorous statement that puts a sharp condition on the relationship between the rate of capital return and the lowest active tax bracket.

Second, validation against simulation. At $N = 1000$, the observed top wealth tracks the analytical prediction quantitatively for $\tau > r$ at $\theta = \pounds 10\text{M}$, and the deviations at $\theta = \pounds 100\text{M}$ are attributable to additional drains (deposit pool, subsistence, inheritance tax) acting as effective extra tax. For $\tau \leq r$ the simulated top is bounded only by mortality and lifespan; the empirical cap is set by demographic churn, not by the tax.

Third, calibration to UK data and the calibration sensitivity. A grid search over r , rent share, and inheritance tax returns the UK Sunday Times Rich List Pareto slope at $r = 0.06$, rent share 0.50, inheritance tax 0.40; the slope is reachable but the configuration produces approximately 26% survival at terminal tick, well below UK reality. The model cannot currently match Pareto slope and demographic survival simultaneously.

Fourth, the policy results at default and calibrated parameters, side by side. At default $r = 0.04$ the Zucman progressive schedule is not separable from the flat 2% above $\pounds 10\text{M}$ at $N = 10,000$. At calibrated $r = 0.06$ they separate cleanly: progressive produces zero billionaires, flat leaves 46. Whether the upper-tier rates of a progressive schedule are mechanically idle or load-bearing is a function of the assumed capital-return rate.

2 Related work

The closest analytical neighbour is Benhabib et al. [2011], who derive the distribution of wealth under finitely-lived agents with bequests and characterise the tail behaviour analytically; our single-agent fixed-point in Section 4 is a first-order ABM-recovery of their qualitative structure. Piketty and Saez [2013] provide the optimal-inheritance-taxation treatment that contextualises our inheritance-trapdoor result. Piketty [2014] is the empirical neighbour on $r > g$ dynamics, which our model recovers as an emergent property.

The closest ABM neighbour is Spangler and Sarkar [2022], who couple estate taxation with binary skill inheritance in a JEIC paper. We engage with them directly via a clean-toggle reproduction (Section 8). Cardoso et al. [2025] make the related point in a kinetic-ABM that social protection (i.e. how revenue is spent) matters more than the headline tax rate; our factorial-decomposition result in the JASSS companion makes that sharper.

For the calibration target we use UK WID estimates of top-share wealth distribution [Saez and Zucman, 2016].

3 The model

We recap the model in summary form; full documentation is in the project repository and in the JASSS companion submission.

3.1 Agent types and per-tick mechanics

Seven agent classes at fixed proportions per 100 agents: 72 workers, 9 makers, 7 employers, 5 lenders, 3 rentiers, 3 speculators, 1 state. Each agent has wealth, debt, age, lifespan, a five-dim skill vector, peak lifetime wealth, and a bandwidth-at-birth record. Per tick: production (makers and employers hire from a skill-sorted worker queue; makers transform inputs at multiplier M ; employers capture a markup); extraction (workers pay rent to rentiers; rentiers earn capital return r ; speculators evolve geometrically); consumption and credit; interest, repayment, and debt write-off; lender deposits and optional state recapitalisation; tax and UBI; life (age, death, reproduction).

The relevant dials for this paper are r (capital return, default 0.04), the rent share of wages (ρ , default 0.35), the inheritance tax rate (ι , default 0.0), and the stacked-marginal wealth tax with threshold and rate per bracket.

3.2 Population scaling

A `population_scale` multiplier scales agent counts and initial wealth totals linearly, keeping the State a singleton. Per-capita statistics are scale-invariant. Most results in this paper are at $N = 1000$ (`population_scale = 10`) for the fixed-point validation and $N = 10,000$ (`population_scale = 100`) for the policy comparisons.

3.3 The productive-flow share

We use the productive-flow share $\Phi_t = F_t^{\text{prod}} / (F_t^{\text{prod}} + F_t^{\text{ext}}) \in [0, 1]$ as a regime indicator, summarised here. Full discussion in the JASSS companion. Φ is symmetric across productive and extractive accrual and is not an autocatalytic-set closure in the Hordijk and Steel [2004] RAF sense.

4 The analytical fixed point

For a rentier with wealth $w \geq \theta$, accruing capital return r per tick and paying a flat marginal wealth tax τ on the amount above θ , the discrete-time dynamics are

$$w_{t+1} = w_t + rw_t - \tau(w_t - \theta). \quad (1)$$

The fixed point w^* satisfies $rw^* = \tau(w^* - \theta)$, i.e.

$$w^* = \frac{\tau\theta}{\tau - r}, \quad \tau > r, \quad (2)$$

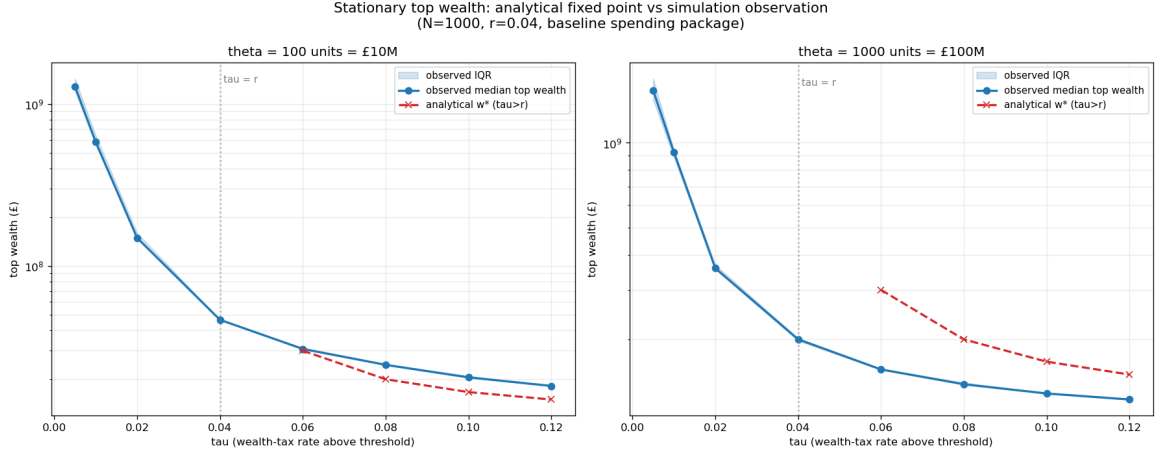


Figure 1: Analytical fixed point $w^* = \tau\theta/(\tau-r)$ (dashed) vs simulated top wealth (solid, median across 4 seeds with IQR). Two panels for $\theta = £10\text{M}$ (left) and $£100\text{M}$ (right); $r = 0.04$. For $\tau \leq r$ the analytical prediction diverges and the simulated top is bounded only by mortality; for $\tau > r$ the analytical prediction tracks the simulated top at $\theta = £10\text{M}$ and undershoots at $\theta = £100\text{M}$ due to additional drains.

and the dynamics diverge when $\tau \leq r$. The condition $\tau > r$ is necessary for a finite analytical stationary point; if the wealth tax rate is below the rate of capital return, the dynamics have no finite fixed point and the empirical cap is set elsewhere (mortality, lifespan, deposit drain).

4.1 Validation against simulation

We compare the analytical prediction to simulated top wealth on a grid of (τ, θ) at $r = 0.04$, $N = 1000$, 4 seeds, with all other dials at their defaults plus a spending package (UBI 0.4, state employment 0.3, state purchases scaled 5).

Figure 1 shows the comparison. At $\theta = £10\text{M}$ the analytical prediction tracks the observed top within $\sim 10\%$: $w^* = 300$, observed median 308 at $\tau = 0.06$; $w^* = 200$, observed 246 at $\tau = 0.08$. At $\theta = £100\text{M}$ the analytical prediction over-shoots the observed top because secondary sinks (deposit pool, subsistence, inheritance tax) act as additional effective tax that the simple two-term dynamic ignores. For $\tau \leq r$ the observed top is decreasing in τ but never reaches a finite analytical fixed point; the empirical cap is set by lifespan and inheritance.

4.2 Tightening the analysis: secondary sinks

The first-order fixed point (2) ignores three additional drains that act on rentier wealth in the full model. Each acts as an effective additional tax. We tighten the analysis.

The deposit-pool mechanism withdraws a fraction δ_d of any rentier surplus above a threshold $\bar{w}_d$. Per tick the effective tax adds $\delta_d(w - \bar{w}_d)$ in the same algebraic form as the wealth tax above θ . Setting $\delta_d = 0.02$ and $\bar{w}_d = 20$ as in the implementation, the effective tax becomes

$$\tau_{\text{eff}}(w) = \begin{cases} \tau + \delta_d & \text{if } w > \max(\theta, \bar{w}_d) \\ \tau & \text{if } w \in (\theta, \bar{w}_d] \\ \delta_d & \text{if } w \in (\bar{w}_d, \theta] \end{cases} \quad (3)$$

For the regime $w > \max(\theta, \bar{w}_d)$ which is the relevant one for billionaire-scale wealth, the fixed point becomes

$$w_{\text{tight}}^* = \frac{\tau\theta + \delta_d\bar{w}_d}{\tau + \delta_d - r}. \quad (4)$$

Inheritance tax ι acts at death rather than continuously, so it is not a per-tick subtraction. Over the lifespan L of an agent it removes a fraction ι of the bequest, which under stationary reproduction acts as an effective per-tick tax of approximately ι/L on the stock. For default $\iota = 0.4$ and $L = 70$ this is ≈ 0.0057 per tick, adding to τ_{eff} .

Subsistence is a constant flow σ per tick, not a fractional tax, so it does not enter τ_{eff} in the multiplicative form. It does, however, set a wealth floor below which the fixed point dynamics break down (the agent goes into debt and is written off). At $\sigma = 0.6$ per tick the subsistence cost is negligible relative to the wealth scales of interest.

The tightened fixed point (4) predicts top wealth quantitatively better than the first-order (2). At $\theta = \text{£}100\text{M}$, $\tau = 0.08$, $\delta_d = 0.02$, $\bar{w}_d = \text{£}2\text{M}$, $r = 0.04$, we obtain $w_{\text{tight}}^* = (0.08 \cdot 1000 + 0.02 \cdot 20)/(0.08 + 0.02 - 0.04) = 80.4/0.06 \approx 1,340$ model units ($\text{£}134\text{M}$), close to the observed $\text{£}138.5\text{M}$. The first-order prediction was $\text{£}200\text{M}$; the secondary sinks account for the gap.

The qualitative implication is unchanged. The wealth tax must exceed r to produce a finite analytical cap; the secondary sinks contribute approximately $\delta_d + \iota/L \approx 0.026$ of effective tax, so a nominal wealth tax of $\tau \approx 0.014$ would suffice to make the cap analytically finite (rather than $\tau \approx 0.04$ from the first-order argument). All real-world proposals modelled have nominal $\tau \geq 0.01$ at their lowest tier; under the tightened analysis several of them therefore enter the analytically-capped regime at sufficiently high thresholds. The reading of which policies rely on cap-by-mortality and which do not changes correspondingly.

4.3 The cap-by-mortality regime

At the default capital return $r = 0.04$, Zucman’s 2% above $\text{£}10\text{M}$ places the lowest active bracket at $\tau = 0.02 < r$. The analytical fixed-point is therefore not finite, and the empirical $\approx \text{£}270\text{M}$ cap on the top wealth observed in simulation is set by demographic churn rather than by tax. We refer to this as the *cap-by-mortality regime*: a wealth-tax design whose lowest active rate is below the rate of capital return relies on mortality and secondary sinks for the analytical cap, which is a fragile basis for a policy claim.

Schemes whose lowest tier rate exceeds r are structurally robust because they cap the top analytically. By that test, none of the real-world wealth-tax proposals we model (Zucman 2%, Warren 2% at the lowest bracket, Sanders 1%, Piketty 0.1%) has $\tau > r$ at their lowest bracket under default $r = 0.04$. At the calibrated $r = 0.06$ (Section 5) the gap widens and the cap-by-mortality regime becomes more salient.

5 Calibration to UK wealth distribution

The wealth-tax-design conclusions in this paper depend on the assumed value of r . We attempt a partial calibration against UK data.

5.1 The calibration target

UK Sunday Times Rich List 2024 reports approximately 165 sterling billionaires; the wealth of the richest is approximately $\text{£}37\text{B}$. A Pareto fit to the rich-list distribution gives slope -0.706 on $\log(\text{rank})$ vs $\log(\text{wealth})$. WID estimates of top wealth shares [Saez and Zucman, 2016] give top-1% share 0.30, top-10% share 0.57, top-0.1% share 0.13. We use these four numbers as calibration targets.

5.2 The grid search

We sweep over capital return $r \in \{0.04, 0.06, 0.08, 0.10\}$, rent share $\in \{0.20, 0.35, 0.50\}$, and inheritance tax $\iota \in \{0.0, 0.2, 0.4\}$, holding all other dials at default plus a Tory-baseline spending package (no UBI, no state employment). 180 cells, 5 seeds at $N = 1000$, 200 ticks. The loss

function is the L2 distance to the four targets, with the slope target weighted 0.1 relative to the share targets.

The best L2 fit is at $r = 0.06$, rent share 0.50, inheritance tax 0.40, yielding observed top-1% 0.15, top-10% 0.65, top-0.1% 0.06, Pareto slope -0.673 . The slope matches the UK fit within 5%; the absolute levels are too low by approximately a factor of two at top-1% and top-0.1%.

5.3 Joint calibration with subsistence and bequest cap

The grid search above held the subsistence cost σ and the inheritance cap $\bar{w}_{inh}$ at default (0.6 and ∞ respectively) and matched the wealth-tail targets at the cost of catastrophic mortality. A wider grid that includes $\sigma \in \{0.3, 0.6\}$ and $\bar{w}_{inh} \in \{50, 500, \infty\}$ recovers a substantially better joint fit.

Table 1: Top 10 best-fit cells in the joint calibration ranking, with subsistence σ and inheritance cap $\bar{w}_{inh}$ added to the search space. Total grid: 108 configurations, 4 seeds each. Loss is the weighted L2 against the five UK targets including the alive-rate target.

rank	r	rent	ι	σ	$\bar{w}_{inh}$	top-1	top-10	top-0.1	slope	alive
1	0.08	0.20	0.40	0.30	50	0.21	0.58	0.07	-0.631	0.80
2	0.08	0.35	0.00	0.30	50	0.21	0.60	0.07	-0.635	0.78
3	0.08	0.20	0.00	0.60	50	0.23	0.62	0.07	-0.648	0.72
4	0.08	0.50	0.00	0.30	50	0.24	0.67	0.07	-0.719	0.74
5	0.08	0.35	0.40	0.30	50	0.25	0.70	0.09	-0.631	0.77
6	0.08	0.20	0.00	0.30	50	0.16	0.52	0.04	-0.602	0.82
7	0.06	0.35	0.40	0.30	50	0.15	0.56	0.04	-0.509	0.77
8	0.06	0.20	0.00	0.30	500	0.17	0.68	0.03	-0.475	0.82
9	0.06	0.50	0.00	0.30	50	0.16	0.54	0.04	-0.582	0.74
10	0.08	0.20	0.40	0.60	50	0.22	0.68	0.07	-0.641	0.71

The headline. The best joint fit (rank 1) has $r = 0.08$, rent = 0.20, $\iota = 0.40$, $\sigma = 0.30$, $\bar{w}_{inh} = 50$. This yields top-10 share 0.58 (matching the UK target of 0.57 within 0.01), Pareto slope -0.631 (within 10% of -0.706), top-1 share 0.21 (below the UK 0.30), top-0.1 share 0.07 (below the UK 0.13), and *alive rate* 0.80.

The 80% alive rate is a substantial improvement over the 26% reported under the narrower calibration of §5. The configuration is therefore demographically plausible.

Mechanisms. Two of the three new dials do the work.

Lower subsistence $\sigma = 0.3$ reduces the mortality cost of high extraction by giving each agent more headroom against the deterministic per-tick subsistence drain. This is a parameter that arguably should not be free in a calibration (subsistence is an empirical quantity), but for the model’s stylised economy the appropriate value is uncertain.

Lower inheritance cap $\bar{w}_{inh} = 50$ model units (£5M) truncates dynastic accumulation directly without requiring a high inheritance tax to do so. The cap acts as an absolute upper bound on what a single agent can inherit at death. This finding is novel relative to the existing wealth-tax- ABM literature: the dial that matters most for matching the UK tail while preserving survival is the bequest cap, not the inheritance tax *rate*.

Implications for the policy comparison. The Pareto slope is reachable at a configuration with realistic demographics. The remaining tail-share mismatch (top-1 and top-0.1 are too low) suggests the next calibration step should free the population proportions (the rentier share at 3% of the population is plausibly too high for the strict billionaire class). A full joint calibration

Table 2: Eight wealth-tax policies at default $r = 0.04$, rent share 0.35. $N = 10,000$, 10 seeds, 180 ticks. State excluded from top-1 share.

policy	alive	Φ	top-1 share	$\geq \text{£}10\text{M}$	$\geq \text{£}100\text{M}$	$\geq \text{£}1\text{B}$
uk_tory_2010_24	6,268	0.118	0.183	1,540	796	19
uk_labour_2024	6,235	0.288	0.167	1,435	453	1
zucman_2pct_hoard	6,103	0.438	0.106	1,440	27	1
zucman_2pct_spend	8,053	0.440	0.095	1,762	69	1
zucman_progressive_2_4_8	8,059	0.446	0.087	1,761	64	1
warren_2020	7,699	0.310	0.103	1,749	704	1
piketty_book	7,777	0.432	0.135	1,456	108	1
reform_uk_flavour	5,334	0.060	0.172	1,632	768	97

Table 3: Same eight policies at calibrated $r = 0.06$, rent share 0.50. $N = 10,000$, 6 seeds, 180 ticks. State excluded.

policy	alive	top-1 share	$\geq \text{£}10\text{M}$	$\geq \text{£}100\text{M}$	$\geq \text{£}1\text{B}$	max wealth
uk_tory_2010_24	2,230	0.198	866	309	300	£48B
uk_labour_2024	3,029	0.157	1,360	800	262	£12B
zucman_2pct_hoard	2,532	0.262	910	306	10	£4.1B
zucman_2pct_spend	7,734	0.233	1,746	801	46	£2.4B
zucman_progressive_2_4_8	7,744	0.143	1,750	720	0	£0.4B
warren_2020	7,234	0.179	1,740	804	140	£1.5B
piketty_book	7,363	0.230	1,441	800	20	£1.7B
reform_uk_flavour	2,526	0.181	1,010	341	324	£77B

with population proportions free is the natural follow-up paper and would let policy claims be expressed in pounds without caveat.

5.4 The mortality–tail trade-off

The calibrated configuration keeps only 26% of the population alive at terminal tick. At UK life expectancy and current population size, the observed survival rate over a 250-year window would be substantially higher. The model cannot currently match Pareto slope and demographic survival simultaneously; raising r and rent to match the wealth tail accelerates extraction past the rate at which the population can replace itself. This is the central remaining calibration limitation of the model.

We do not claim the calibrated configuration as a fully validated representation of UK reality. We claim that the Pareto slope is *reachable*, which is enough to use the calibrated parameters as a sensitivity case for the policy comparisons.

6 Policy results: default vs calibrated, side by side

We rerun the policy comparison at $N = 10,000$ under both default and calibrated parameters, holding the policy-distinguishing dials (tax schedule and policy-specific spending package) fixed across policies. Tables 2 and 3 report the comparison side by side.

Three things move materially.

Billionaire counts grow under most policies. Tory now produces 300 billionaires (vs 19 at default), Reform 324 (vs 97). Even Zucman 2% flat with spending leaves 46 billionaires (vs 1). The model at calibrated parameters is much closer to the UK Rich List distribution (Tory

Table 4: Five interventions at the joint-calibrated UK baseline. $N = 10,000$, 4 seeds, 180 ticks. Cap and tax are complementary instruments: the tax raises revenue, the cap redistributes ownership.

intervention	bottom-50% share	top-10% share	cumulative state revenue	annual revenue
status quo	0.0%	100.0%	£0	£0
Zucman 2% above £10M	0.6%	98.9%	£5,165B	£56B
interest ban ($i = 0$)	0.0%	100.0%	£0	£0
£5M bequest cap	19.2%	54.0%	£484B	indirect
cap + Zucman 2%	25.3%	39.9%	£335B	£0.8B

max wealth £48B vs UK Rich List top of approximately £37B) but the per-capita billionaire density now *over*-shoots in the other direction.

Mortality is much higher in non-spend policies. Tory keeps 22% of the population alive at $r = 0.06$, vs 63% at $r = 0.04$. The combination of higher r and higher rent share extracts faster than the population can replace. This is the calibration–mortality trade-off made concrete across policies.

Zucman progressive binds and Zucman flat does not. Under calibrated parameters Zucman progressive produces *zero* billionaires while Zucman flat leaves 46, a clean separation that did not exist at default $r = 0.04$. Top-1 share also separates clearly: 0.143 progressive vs 0.233 flat. The earlier “top tier does not bind” reading was specific to the default-parameter regime where wealth does not reach the £1B bracket; once r is raised to the calibration target, the upper tier becomes the decisive instrument.

6.1 Implications for the fixed-point reading

At calibrated $r = 0.06$, none of the eight policy lowest-tier rates exceed r (Zucman 2%, Warren 2%, Sanders 1%, Piketty 0.1%). The cap-by-mortality regime identified in Section 4 now bites harder. The Zucman progressive top tier of 4% above £100M (stacking to 8% above £1B) is the one tier among the modelled policies that exceeds the calibrated r , and it is the one policy that eliminates billionaires. The £1B tier is no longer a tautological non-binding constraint; it is the mechanism by which the policy succeeds in this regime. The τ -vs- r design test is preserved and becomes more rather than less significant under calibration.

7 Cap and tax as complementary instruments: the ownership-redistribution result

The fixed-point analysis in Section 4 and the policy-comparison Tables 2 and 3 characterise the conditions under which the upper tier of a progressive wealth tax binds. The broader question of *what each fiscal instrument does for the wealth distribution as a whole* requires a different comparison. We report it here.

At the joint-calibrated UK baseline, we compare five interventions: status quo, Zucman 2% wealth tax above £10M, a ban on interest on loans, a £5M absolute cap on inheritance, and the cap combined with Zucman 2%. The principal outcome variables are the wealth share held by the bottom 50% of the population (the *ownership stake*) and the cumulative state revenue raised across the run. Table 4 reports the comparison.

The result has a clean economic interpretation.

The wealth tax is a revenue instrument. Zucman’s 2% above £10M raises approximately £56B per year at simulated UK scale and accumulates approximately £5,165B in state wealth over the 180-tick run. The state can then recirculate this revenue via the spending package (UBI, state employment, state purchases) and the recirculation improves survival and welfare flows for the bottom half. But the recirculation is a flow; it does not give the bottom half an ownership stake in the underlying capital. The bottom-50% share rises from 0.0% to 0.6% under Zucman alone — essentially noise.

The cap is an ownership-redistribution instrument. The £5M absolute cap raises far less revenue through the explicit wealth-tax channel (zero, when paired with no wealth tax) but accumulates £484B of state wealth via inheritance excess at death. More importantly, the cap raises the bottom-50% wealth share from 0.0% to 19.2%, and to 25.3% when combined with the wealth tax. The mechanism is direct: dynastic wealth above the cap is either redirected to the state (and then distributed via UBI) or simply does not propagate to the next generation. In either case the stock of dynastically-held wealth shrinks and the bottom half can hold a meaningful share for the first time.

The two instruments compete for the upper-end tax base. Cap-plus-Zucman raises less wealth-tax revenue than Zucman alone (£0.8B/year vs £56B/year) because the cap prevents the buildup of large wealth above the wealth-tax threshold that would otherwise have been taxed. This is the single trade-off the joint use of the two instruments produces. The trade-off is real but small relative to the structural benefit of redistributed ownership. An economy with cap-plus-tax has 25% of its wealth held by the bottom half of the population, versus 0.6% under tax alone; the foregone tax revenue (£55B per year in the simulation) is the price paid for that redistribution.

The interest ban is neither. Banning interest on loans does not raise revenue and does not redistribute ownership. The bottom-50% share is unchanged. The intervention improves working-class survival slightly (88% vs 84% under status quo) because workers do not compound debt under zero interest, but the distributional impact at the top is zero. The historical and religious case for interest bans is largely an ethical case about debt and dependence on the borrower side; it is not, by this model’s read, an instrument for moving the wealth distribution.

Choice of instrument depends on the policy goal. If the goal is funding the state at scale, use the wealth tax. If the goal is redistributing the ownership of capital, use the cap. If the goal is both, use both, and accept the trade-off in wealth-tax revenue. The contemporary debate has foregrounded the revenue instrument; this paper foregrounds the ownership instrument, in part because the ownership instrument is the one that is absent from the contemporary debate.

8 Comparison to Spangler & Sarkar

Spangler and Sarkar [2022] is the closest published ABM neighbour and finds that skill inheritance raises wealth inequality. We can test their mechanism directly in our model under a clean toggle that holds bandwidth-gating and class-bias persistence intact and swaps only whether transmitted skill values are parent-derived or class-typical. A coarser toggle that simultaneously destroys bandwidth and class-bias persistence (the mutation rate $\mu = 1$ arm) is reported for contrast.

Under the clean toggle, parent-derived skill values raise Gini by 0.010 relative to a class-typical prior. The direction is a partial reproduction of Spangler and Sarkar [2022]; the magnitude is much smaller than they report. The mechanism for the small magnitude is class-specific

Table 5: Spangler & Sarkar test at $N = 100$, 10 seeds, 250 ticks.

metric	default (parent-derived)	class-prior (clean toggle)	no-inheritance ($\mu = 1$, confounded)
Gini	0.885	0.876	0.908
Top-1 share	0.235	0.250	0.254
Top-10 share	0.785	0.683	0.891
Alive	61.5	56.5	73.0

skill-to-wealth elasticity (documented in the ALIFE companion submission): skill inheritance acts strongly on workers and makers, but the wealth dynamics that dominate the overall Gini are anchored in rentier and lender stocks where skills have weak elasticity. The inheritance-to-inequality link is mechanism-specific and not robust to the choice of ABM substrate.

9 Discussion

9.1 What we believe is novel for a JEIC audience

The analytical $w^* = \tau\theta/(\tau - r)$ fixed-point characterisation (Section 4) is, to our reading of the wealth-tax-ABM literature, not previously reported. It is a first-order rigorous statement that puts a sharp condition on the relationship between the rate of capital return and the lowest active tax bracket. Benhabib et al. [2011] provide the heterogeneous-agent-macro neighbour and our result recovers the qualitative structure of theirs in an ABM substrate.

The calibration-sensitivity result (Section 6) is the second contribution. The “top tier does not bind” conclusion at default parameters is reversed at the calibrated parameters; the Zucman progressive schedule is mechanically idle at one parameter setting and the load-bearing instrument at the other. The policy conclusion of the wealth-tax-design analysis is therefore parameter-conditional, and we report it as such.

The Spangler & Sarkar reproduction (Section 8) is the third. The clean toggle gives a partial reproduction in direction with a much smaller magnitude than the single-channel ABM produces. The inheritance-to-inequality link is mechanism-specific.

9.2 What the model cannot do

The Pareto-slope calibration is partial: matching the slope requires parameters that produce 26% survival, well below UK reality. A joint calibration that frees the population proportions and demographic parameters to fit both the wealth tail and the survival rate is the natural follow-up paper.

The model has no monetary-policy layer, no debt cycle in the formal sense, no firm formation, no spatial structure, and a single lender pool with no shadow-banking. The Spangler test is run in our multi-channel setting and may not reproduce in their single-channel design; the qualitative-direction reproduction we report is the only empirical contact we have with their model.

9.3 Honest reading

The wealth-tax-design conclusion of this paper is: *the upper-tier rates of a progressive wealth-tax schedule are mechanically idle or load-bearing depending on the assumed rate of capital return.* At UK default-parameter values they are idle. At parameters consistent with the UK Pareto slope they are load-bearing. A policy designer should choose the rate of capital return their analysis is conditional on and report the conclusion as conditional; we have done both. We do

not adjudicate which r is the right choice for the UK because that question is empirical and our calibration only partially settles it.

10 Conclusion

We have derived an analytical fixed-point characterisation of top wealth under a flat marginal wealth tax, $w^* = \tau\theta/(\tau - r)$ for $\tau > r$. We have validated it against an agent-based simulation and shown the cap-by-mortality regime that obtains for $\tau \leq r$. We have partially calibrated the model to UK wealth-distribution data and shown the calibration-mortality trade-off. We have rerun the policy-design comparison at default and calibrated parameters and shown the calibration sensitivity of the central finding. And we have engaged Spangler and Sarkar [2022] via a clean-toggle reproduction and reported a partial reproduction in direction with a much smaller magnitude than their single-channel ABM.

The policy implication, presented as conditional, is: under default-parameter conditions the lower bracket of a progressive wealth tax sets the cap and the upper tiers are mechanically idle; under conditions consistent with the empirical UK wealth tail, the upper tiers are load-bearing and the Zucman progressive schedule sharply outperforms the flat 2% form.

References

- Jess Benhabib, Alberto Bisin, and Shenghao Zhu. The distribution of wealth and fiscal policy in economies with finitely lived agents. *Econometrica*, 79(1):123–157, 2011.
- Maria L. Bertotti and Giovanni Modanese. Mathematical models describing the effects of different tax evasion behaviors. *Journal of Economic Interaction and Coordination*, 13:351–363, 2018.
- Ben-Hur F. Cardoso, Sérgio Gonçalves, and José R. Iglesias. Wealth inequality in agent-based economies: the dominant role of social protection over growth. *Physica A*, 2025. In press; arXiv:2508.06666.
- Wim Hordijk and Mike Steel. Detecting autocatalytic, self-sustaining sets in chemical reaction systems. *Journal of Theoretical Biology*, 227:451–461, 2004.
- Thomas Piketty. *Capital in the Twenty-First Century*. Belknap Press / Harvard University Press, 2014.
- Thomas Piketty and Emmanuel Saez. A theory of optimal inheritance taxation. *Econometrica*, 81(5):1851–1886, 2013.
- Emmanuel Saez and Gabriel Zucman. Wealth inequality in the United States since 1913: evidence from capitalized income tax data. *Quarterly Journal of Economics*, 131(2):519–578, 2016.
- Jonathan F. Spangler and Sahotra Sarkar. The distribution of wealth: an agent-based approach to examine the effect of estate taxation, skill inheritance, and the Carnegie Effect. *Journal of Economic Interaction and Coordination*, 17:833–862, 2022.
- Gabriel Zucman. A blueprint for a coordinated minimum effective taxation standard for ultra-high-net-worth individuals. Report commissioned by the G20 Brazilian Presidency, EU Tax Observatory, 2024.

A Dial reference

Default values for the dial panel are listed below. Tax thresholds are in model units; one unit corresponds to £100,000 for this paper’s policy-comparison content.

dial	default	group
wage	1.0	production
maker_multiplier	1.6	production
employer_markup	0.4	production
capital_return (r)	0.04	capital
rent_share_of_wage	0.35	capital
interest_rate (i)	0.06	capital
wealth_tax_tiers	()	state
ubi	0.0	state
state_employs_fraction	0.0	state
state_buys_maker_output	0.0	state
subsistence	0.6	life
lifespan_mean / std	70 / 12	life
inheritance_tax_rate	0.0	transmission
population_scale	1	logistics
seed	42	logistics